UNIT 1 · TOPIC 1.3

Public Network Sniffing

Quiz

Instructions: This quiz has 18 multiple-choice questions, 1 point each. Circle the letter of the single best answer for each question.

Name: **Date:** **Class:**

1. Which term describes a rogue wireless access point that broadcasts the same network name (SSID) as a legitimate nearby network in order to trick users into connecting to it?

- ○ A. Evil twin attack
- ○ B. Deauthentication attack
- ○ C. ARP poisoning attack
- ○ D. Replay attack

2. A person sets up an unsecured wireless hotspot in a coffee shop without authorization, intending to intercept login credentials from nearby customers for personal financial gain. Which category of adversary does this most accurately describe?

- ○ A. Nation-state threat actor
- ○ B. Opportunistic cybercriminal
- ○ C. Hacktivist
- ○ D. Insider threat

3. In which type of wireless attack does an adversary secretly position themselves between a user and a legitimate access point, intercepting and potentially altering communications without either party's knowledge?

- ○ A. Brute force attack
- ○ B. SQL injection
- ○ C. Denial of service (DoS) attack
- ○ D. Man-in-the-middle (MitM) attack

4. A student at a transit hub connects to a network labelled "Free_Transit_WiFi" and is redirected to a captive portal that requests her social media credentials. She enters them and gains internet access, but later discovers her social media password has been changed by an unknown party. Which combination of wireless attack techniques does this scenario most precisely illustrate?

- ○ A. Evil twin access point combined with credential harvesting via a fake captive portal
- ○ B. Deauthentication attack followed by a replay attack
- ○ C. ARP poisoning followed by session hijacking at the router
- ○ D. Bluejacking combined with bluesnarfing over Bluetooth

5. A user submits their online banking password over an unencrypted HTTP connection while connected to a public Wi-Fi network. Which statement most accurately describes why this creates a cybersecurity risk?

- ○ A. The banking server's firewall will block the unencrypted request before it reaches the server
- ○ B. WPA2 encryption on the access point protects the credentials from any third-party interception
- ○ C. The credentials are transmitted as readable plaintext and can be captured by any device monitoring the same network segment

○ D. The browser automatically hashes the password before sending it over HTTP

6. An adversary monitors unencrypted traffic at an airport lounge, collects plaintext email addresses and passwords, and tests those credentials against a banking website. Many stolen passwords work on the banking site because users reused them across services. How does this attack sequence most accurately illustrate the risk generated by credential theft on public networks?

○ A. It compromises network infrastructure integrity but does not affect individual account confidentiality

○ B. It enables credential stuffing across multiple services, creating account-takeover risk that extends far beyond the original breach point

○ C. It primarily affects the availability of the airport network rather than the confidentiality of user data

○ D. It introduces malware onto victims' devices via the wireless interface, enabling persistent unauthorized access

7. In Scenario 1C, the victim joins "Guest Wi-Fi" instead of "Sunshine Coffee Wi-Fi," enters streaming platform credentials into a fake captive portal, and is later locked out of her music account. Which sequence of attack stages most completely and accurately characterizes what occurred?

○ A. Reconnaissance → Exploitation → Exfiltration → Privilege escalation

○ B. Network spoofing → Credential harvesting → Credential stuffing → Account takeover

○ C. Phishing email → Malware installation → Lateral movement → Persistence

○ D. ARP poisoning → Session hijacking → Data exfiltration → Denial of service

8. Evaluate the following claim: "Setting up an evil twin access point requires specialized hardware and advanced networking expertise, making it an unlikely threat from opportunistic adversaries." Which response most accurately critiques this assertion?

○ A. The claim is correct; evil twin attacks require custom firmware unavailable to non-expert attackers

○ B. The claim is partially correct; initial setup is simple, but avoiding detection requires advanced skills

○ C. The claim is incorrect; commodity tools and consumer-grade hardware reduce the technical barrier for evil twin attacks to a level accessible to low-skill adversaries

○ D. The claim is incorrect; evil twin attacks are only feasible for nation-state actors because of the cost of the required equipment

9. A security analyst reviews an incident where hotel guests joined a rogue access point, authenticated through a fake captive portal, and had multiple linked accounts subsequently compromised. Which combination of threat characteristics most completely describes the attack surface that was exploited?

○ A. Local access vector · medium complexity · software vulnerability exploitation · persistence via installed malware

○ B. Network access vector · high attack complexity · cryptographic weakness in WPA2 · single-service account impact

○ C. Physical proximity vector · low attack complexity · credential-based authentication weakness · multi-service credential reuse

○ D. Adjacent access vector · high complexity · zero-day exploit · privilege escalation to administrator

10. Which of the following security controls most directly protects data transmitted over a public Wi-Fi network by creating an encrypted tunnel for all outbound traffic?

○ A. A strong, unique account password

○ B. Multi-factor authentication (MFA)

○ C. A Virtual Private Network (VPN)

○ D. A device screen lock PIN

11. What security property does HTTPS provide that plain HTTP does not, making HTTPS preferable when transmitting sensitive information on any network?

- ○ A. It hides the user's IP address from the destination web server
- ○ B. It prevents the web server from creating logs of user activity
- ○ C. It prevents third parties from identifying the domain names the user visits
- ○ D. It encrypts data transmitted between the user's browser and the web server

12. Which of the following actions best reduces the risk of data interception when a person must use a public Wi-Fi network?

- ○ A. Using HTTPS websites and activating a VPN for all connections
- ○ B. Disabling Bluetooth and connecting to any available open network
- ○ C. Selecting the network with the strongest signal to ensure the fastest connection
- ○ D. Logging out of sensitive accounts only after returning home from the public location

13. A user connects to an untrusted public Wi-Fi access point and activates a VPN before accessing any websites. How does the VPN specifically mitigate the risk of data interception on this network?

- ○ A. The VPN authenticates the access point's certificate, preventing users from connecting to rogue APs
- ○ B. The VPN replaces WPA2 wireless encryption, making the underlying access point's security configuration irrelevant
- ○ C. The VPN blocks all unsolicited inbound connections, functioning as a host-based firewall on the public network
- ○ D. The VPN encrypts all outbound traffic before it leaves the device, so any data intercepted on the public network remains unreadable to the adversary

14. A student needs to submit a graded assignment through her school's online portal while sitting in a public library. She wants the combination of controls that best protects her credentials and submission. Which set of actions provides the most complete risk mitigation for this situation?

- ○ A. Confirm the correct library Wi-Fi name with a staff member, use only HTTPS connections, and activate a VPN before logging in
- ○ B. Connect to any available network and rely on the school portal's password complexity requirements
- ○ C. Select the network with the most connected users and enable file sharing for faster upload speeds
- ○ D. Use the library's Ethernet port if a VPN is unavailable, without requiring HTTPS

15. A user's smartphone is configured to automatically rejoin any Wi-Fi network it has previously connected to. Why is disabling this auto-connect feature considered a security control for users who frequently use public networks?

- ○ A. It reduces battery consumption, which prevents the device's location from being tracked via signal-strength analysis
- ○ B. It prevents the device from silently joining a rogue access point that broadcasts an SSID matching a previously trusted network
- ○ C. It stops adversaries from sending deauthentication frames that force the device off its current legitimate connection
- ○ D. It encrypts the list of stored Wi-Fi network names held in the device's configuration

16. In Scenario 1C, the victim could have applied security controls at three stages: before joining the network, while connected, and after the session. Which layered sequence of controls would have been most effective in preventing the credential theft?

- ○ A. Before: connect to the highest-signal network; During: use HTTP for faster speeds; After: change passwords when the service prompts
- ○ B. Before: disable VPN to allow captive portal access; During: rely on MFA alone; After: review browser history for suspicious sites
- ○ C. Before: enable auto-connect for convenience; During: use OAuth login for speed; After: run an antivirus scan
- ○ D. Before: verify the correct SSID with coffee shop staff; During: decline OAuth login on any captive portal and activate a VPN; After: monitor linked accounts for unauthorized changes

17. A network policy analyst evaluates a proposal to replace open captive portals at public venues with 802.1X port-based access control. Which argument most accurately supports this change as a risk mitigation measure against the attack type illustrated in Scenario 1C?

- ○ A. 802.1X requires users to create unique venue-specific credentials, increasing password diversity across services
- ○ B. 802.1X encrypts the SSID broadcast itself, preventing adversaries from cloning the network name
- ○ C. 802.1X eliminates browser-based captive portals, removing the credential-harvesting attack surface they represent
- ○ D. 802.1X mandates multi-factor authentication, making any account takeover impossible once deployed

18. A company policy requires all employees to use a VPN on public networks, but includes an exception allowing them to temporarily disable the VPN to complete captive portal login. A security auditor flags this exception as a significant vulnerability. Which explanation most precisely justifies the auditor's concern?

- ○ A. The VPN actively inspects captive portal TLS certificates, and disabling it prevents certificate validation, increasing man-in-the-middle exposure at the application layer
- ○ B. Disabling the VPN prevents automatic reconnection after portal authentication, leaving all subsequent session traffic unencrypted indefinitely
- ○ C. Disabling the VPN deactivates WPA2 on the device, leaving the wireless channel permanently unencrypted for the rest of the session
- ○ D. Disabling the VPN to authenticate through a captive portal creates an unprotected window during which pre-tunnel traffic, including captive portal credentials, can be intercepted by an adversary controlling the access point